



Original Research

Historic trends and future projections of the prevalence of adult excess weight in Scotland, 2003 to 2040: A modelling study

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ABSTRACT

Objectives: The prevalence of excess weight in Scotland is higher than other UK nations and amongst the highest when compared with European Union countries. We aimed to use historic data to project estimates of the prevalence of excess weight in Scotland using Bayesian age-period-cohort modelling.

Study design: A Bayesian age-period-cohort modelling study.

Methods: Interviewer-validated height and weight measurements were obtained from the Scottish Health Survey (2003–2019) to calculate body mass index (BMI). Relevant socio-demographic attributes were sourced to contribute to modelling and results stratification. Study outcomes were defined as overweight (BMI ≥ 25 kg/m²) and obesity (BMI ≥ 30 kg/m²). We estimated historic trends in study outcomes by sex and age-group, and projected future trends to 2040, using Bayesian hierarchical age-period-cohort modelling.

Results: In 2040, we estimate 3.3 million adult cases of overweight, of which 1.6 million are cases of obesity. Projections were more pronounced for obesity, compared to overweight, with an additional 66,000 male cases and 175,000 female cases projected between 2019 and 2040. Between 2003 and 2019, the proportion of male and female cases of both overweight and obesity aged 65 years and above have increased, a trend projected to further intensify as we move towards 2040.

Conclusions: Left unaddressed, we estimate a substantial increase in adult excess weight in Scotland by 2040, particularly for females, compounded by increases in obesity and in the proportion of older cases. These findings are a warning signal of future adverse population health impacts and healthcare service sustainability pressures. Projections are not inevitable and underscore the need to accelerate progress on implementing preventative measures to address the food environment, and on further development of weight management and support services, to improve Scotland's population health.

1. Introduction

Recent estimates indicate around two-thirds of Scottish adults are overweight, and one-third are living with obesity, rates disproportionately higher compared to other UK nations and among the highest compared with European Union countries, with higher rates persisting

for Scotland's most deprived areas.^{1–3} The most widespread metric used for determining whether an individual's weight is higher than recommended is a person's body mass index (BMI). Excess weight is associated with increased risk of a wide range of health conditions, including mortality, from: diabetes and kidney diseases; cardiovascular diseases; cancers; musculoskeletal disorders; respiratory infections and chronic

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diseases; digestive diseases; and neurological disorders.⁴ Approximately 10 % of all-cause health loss in Scotland is attributable to excess weight, which also presents substantial demands and pressures on health and social care services.⁵ The annual cost of obesity in Scotland is estimated at £5.3 billion, with the largest share from associated reductions in quality of life.⁶ These costs do not consider costs attributable to overweight excluding obesity, which is highly prevalent and remains associated with significant adverse health outcomes, indicating that the cost of excess weight is likely to be considerably more.

For more than two decades, the Scottish Government has sought to address obesity with strategies seeking to improve lifestyle-focused education in addition to recognising the important role of food providers to support healthy eating.⁷ The most recent strategy, the 2018 Diet and Healthy Weight Delivery Plan, further emphasised the focus on the role of industry and other providers of food with improvements to food environments one of the key underpinning outcomes sought.⁸ This recognises that individual choices are shaped by broader structural factors such as affordability, availability, and marketing of unhealthy food. Despite these efforts, obesity prevalence has remained high, and inequalities have widened, prompting the recently launched Population Health Framework to identify obesity — and the systems that drive it — as a key priority for action in the coming years.⁹ A new Diet and Health Weight Implementation Plan is in development as part of the commitments in the Population Health Framework.

Scottish Burden of Disease study findings illustrate that, if left unaddressed, changes in population and ageing would be expected to significantly increase disease burdens by around 21 % by the early 2040s, placing intense demands on the healthcare system.¹⁰ This highlights the importance of utilising epidemiological methods which allow us to anticipate future trends of key public health concerns, such as high BMI. Scotland has a longstanding annual survey which collects validated data on the BMI of respondents, presenting a key opportunity to harness this data to undertake age-period-cohort (APC) analysis.¹¹ APC analysis allow us to separate, and project trends of, effects related to: age on excess weight; how excess weight develops over time; and, the difference in the risk of excess weight in successive birth cohorts. This approach is advantageous since period and cohort effects serve as proxies for events such as changes in the fundamental determinants of health, and public health and medical interventions, which are often difficult to measure directly.

We aimed to use historic data to identify APC effects and integrate them into projected estimates of adult prevalent cases of excess weight for males and females in Scotland.

2. Methods

2.1. Study design

The study setting was Scotland, a country with a population of 5.5 million residents, of which 4.6 million are adults aged 16 years and above.¹² The sample was restricted to adults, and the sampling period for assessing historic APC effects in the prevalence of excess weight was 2003–2019 with a future projection period up to 2040. This study is reported in adherence with the STROBE statement ([appendix p 17](#)).

2.2. Data

We sourced data from the UK Data Service for 13 cross-sectional waves of the Scottish Health Survey (2003, 2008 to 2019 inclusive).¹³ The Scottish Health Survey provides a detailed picture of population health through sampling private households using stratified cluster probability sampling of the non-institutionalised residential population, which has been described in further detail elsewhere.¹⁴ Prior to 2008, the Scottish Health Survey was conducted intermittently with previous surveys in 1995 and 1998 restricted to sampling one adult per household. In 2003, this restriction was removed and from 2008 the survey

was carried out annually with smaller sample sizes. Interviewer-validated height and weight measurements were sourced to calculate BMI ($BMI = \frac{Weight(kg)}{Height(m)^2}$) and study outcomes were defined as

overweight ($BMI \geq 25 \text{ kg/m}^2$) and obesity ($BMI \geq 30 \text{ kg/m}^2$).

The age at survey date, sex, and area-level deprivation quintile defined by the Scottish Index of Multiple Deprivation (SIMD) was sourced for survey respondents. SIMD is a small-area index of multiple deprivation based on approximately 7,000 data zones, incorporating pooled and weighted indicator data from seven domains: employment; income; crime; housing; health; education; and geographical access.¹⁵ The resulting composite SIMD ranking for each data zone is subsequently converted into quintiles, where the first reflects the 20 % most deprived areas, with the fifth reflecting the 20 % least deprived areas. The age at survey date was used to establish the birth year for each respondent. Explanatory variables into APC models were sex, age at survey date, birth year, survey year, and SIMD quintile. Survey weights were obtained and used to ensure that estimates were adjusted to account for variations in sampling and survey responses across each of the survey waves.

Mid-year Scottish annual population estimates for 2003 to 2019 were sourced from National Records of Scotland (NRS) by sex, single year of age, and SIMD quintile.¹² Annual population projections up to 2040 were sourced from NRS by sex and single year of age.¹⁶

2.3. Analyses

We used hierarchical APC (HAPC) modelling to estimate the prevalence of overweight and obesity using two separate HAPC models. HAPC models allow for a mixture of fixed and random effects, whereas there is a linear dependency between age, period, and cohort effects in classical APC models.¹⁷ Further analysis was undertaken to explore the impact of the possible loss of information from modelling dichotomised outcomes by alternatively modelling the continuous variable of population-level BMI and comparing their results.

We applied second-order random walk priors to age, period, and cohort effects.¹⁸ Model selections were based on the Deviance Information Criteria (DIC) and Watanabe–Akaike Information Criterion (WAIC). The fit and predictive accuracy of models were assessed by fitting the model to repeated cross-sectional sample surveys. The proportion of variation in the outcome that was explained by the independent variables (R^2) and the root mean squared error (RMSE) were estimated for different priors and models.

Estimates are described using the median, and 95 % credibility intervals (CrI) around the median, constructed based on the 2.5th and 97.5th percentile of the posterior distribution. Results relate to absolute numbers of prevalent cases of overweight and obesity, as our motivation was to consider how these estimates could impact population-level needs and demands in the future. This means that our results do not control for differences in underlying population structure. Utilising the strengths of applying Bayesian HAPC models, we estimated the probability of scenarios of increased prevalence for overweight and obesity by 2030 and 2040, separately for males and females.¹⁹ The thresholds used were increases greater than: 0 %, 5 %, 10 %, 25 %, 50 %, and 100 %, where a 100 % increase would be equivalent to double the reference category.

All statistical analyses were performed in R 4.2.0.²⁰ HAPC models were fitted using the Integrated Nested Laplace Approximation (INLA) package (Version 22.05.07).²¹

3. Results

3.1. Study sample

The overall sample of adults aged 16 years and above was 72,542

participants across all 13 cross-sectional waves of the Scottish Health Survey ([appendix p 4](#)). Generally, the sample size decreased over time from 2003 to 2019. The median age of participants ranged from 50 to 54 years, with a higher proportion of female participants. The sex distribution was consistent across survey waves, ranging from 55 to 57 % female respondents. The proportion of respondents from Scotland's least deprived areas slightly increased over time from 2003 to 2019, whilst the proportion of respondents from Scotland's most deprived areas slightly decreased over time, although there were some annual fluctuations in this long-term trend.

3.2. Contribution of factors to the prevalence of excess weight

The odds of overweight were higher for males (Females vs Males, Odds Ratio (OR) = 0.69; 95 % CrI = 0.67 to 0.72), whereas the odds of obesity were higher for females (OR = 1.07; 95 % CrI = 1.03 to 1.11) ([Table 1](#)). The odds of both overweight and obesity increased by increasing levels of area-level deprivation. Odds increased linearly for obesity from 31 % higher for SIMD 4 (SIMD 4 vs SIMD 5, OR = 1.31; 95 % CrI = 1.23 to 1.39) to 87 % higher for the most deprived areas (SIMD 1 vs SIMD 5 OR = 1.87; 95 % CrI = 1.76 to 1.98). This trend was similar for overweight, although at a slightly lower level, and the highest odds of overweight were estimated for those residing in SIMD 2; the second most deprived area-level deprivation quintile (SIMD 2 vs SIMD 5 OR = 1.45; 95 % CrI = 1.38 to 1.53). The odds of both overweight and obesity

by age followed a similar pattern and were highest for the ages of 35–64 years, and then started to reduce from the ages of 65 years and above but remained higher than the under 25 years reference group until the ages of 85 years and above. The odds of overweight and obesity reduced for successive birth cohorts, with the lowest odds estimated for more recent birth cohorts. On the other hand, the odds of overweight and obesity increased by each period with the highest odds being estimated for the most recent periods (2019 vs 2003, Overweight OR = 1.54; 95 % CrI = 1.52 to 1.57; Obesity OR = 1.59; 95 % CrI = 1.56 to 1.62). Visualisations of the estimated fixed and random effects for models are presented in the [appendix \(pp 5–9\)](#).

3.3. Trends in the prevalence of excess weight

Estimated cases of overweight and obesity for males increased by 14 % and 38 % respectively from 2003 to 2019, with the fastest rate of increase occurring between 2003 and 2010 ([Fig. 1](#)). For females, we found that the estimated cases of overweight and obesity increased by 13 % and 25 % respectively from 2003 to 2019, with the rate of increase broadly consistent across the full study sampling period.

In 2040, we estimate 3.3 million adult cases of overweight, of which 1.6 million are cases of obesity ([appendix pp 13–16](#)). Projections of overweight cases for males are estimated to increase by 3 % from 2019 to 2030; an increase of approximately 46,000. However, projected cases of overweight for males are estimated to slowly decline after a peak,

Table 1

Odds ratios for excess weight in Scotland based on Bayesian age-period-cohort model of sample data by study outcome.

		Overweight			Obesity		
		OR	95 % CrI		OR	95 % CrI	
			Lower	Upper		Lower	Upper
Sex	(Intercept)	1.53	1.41	1.65	0.21	0.20	0.23
	Male	Reference group			Reference group		
SIMD	Female	0.69	0.67	0.72	1.07	1.03	1.11
	SIMD 5 [Least deprived]	Reference group			Reference group		
SIMD	SIMD 4	1.23	1.17	1.30	1.31	1.23	1.39
	SIMD 3	1.34	1.27	1.42	1.48	1.39	1.56
	SIMD 2	1.45	1.38	1.53	1.73	1.63	1.83
	SIMD 1 [Most deprived]	1.37	1.30	1.45	1.87	1.76	1.98
Age-group	Under 25 years	Reference group			Reference group		
	25–34 years	1.68	1.63	1.74	1.53	1.48	1.61
	35–44 years	2.20	2.08	2.35	2.05	1.91	2.23
	45–54 years	2.35	2.18	2.56	2.04	1.86	2.27
	55–64 years	2.33	2.16	2.54	2.06	1.89	2.28
	65–74 years	2.03	1.93	2.16	1.79	1.68	1.94
	75–84 years	1.48	1.45	1.52	1.18	1.15	1.23
	85 years and above	0.97	0.90	1.04	0.60	0.55	0.65
Birth cohort	Before 1920	Reference group			Reference group		
	1920 to 1929	1.02	0.93	1.18	0.90	0.84	1.03
	1930 to 1939	1.03	0.90	1.25	0.81	0.73	0.98
	1940 to 1949	0.95	0.81	1.19	0.72	0.64	0.90
	1950 to 1959	0.79	0.66	1.02	0.64	0.56	0.82
	1960 to 1969	0.66	0.55	0.86	0.56	0.48	0.74
	1970 to 1979	0.55	0.46	0.70	0.49	0.42	0.64
	1980 to 1989	0.44	0.38	0.55	0.43	0.37	0.54
	1990 to 1999	0.33	0.29	0.40	0.37	0.33	0.46
	2000 and above	0.24	0.22	0.29	0.33	0.30	0.37
Period	2003	Reference group			Reference group		
	2008	1.19	1.16	1.22	1.21	1.17	1.25
	2009	1.21	1.17	1.26	1.25	1.20	1.30
	2010	1.24	1.19	1.29	1.28	1.23	1.34
	2011	1.26	1.21	1.32	1.31	1.25	1.38
	2012	1.29	1.23	1.35	1.34	1.28	1.41
	2013	1.31	1.26	1.37	1.37	1.31	1.44
	2014	1.34	1.29	1.40	1.41	1.34	1.48
	2015	1.37	1.32	1.43	1.44	1.38	1.51
	2016	1.41	1.36	1.47	1.48	1.42	1.55
	2017	1.45	1.40	1.51	1.51	1.46	1.58
	2018	1.50	1.45	1.54	1.55	1.50	1.60
	2019	1.54	1.52	1.57	1.59	1.56	1.62

OR denotes 'Odds ratio'; 95 % CrI denotes '95 % Credible Interval'.

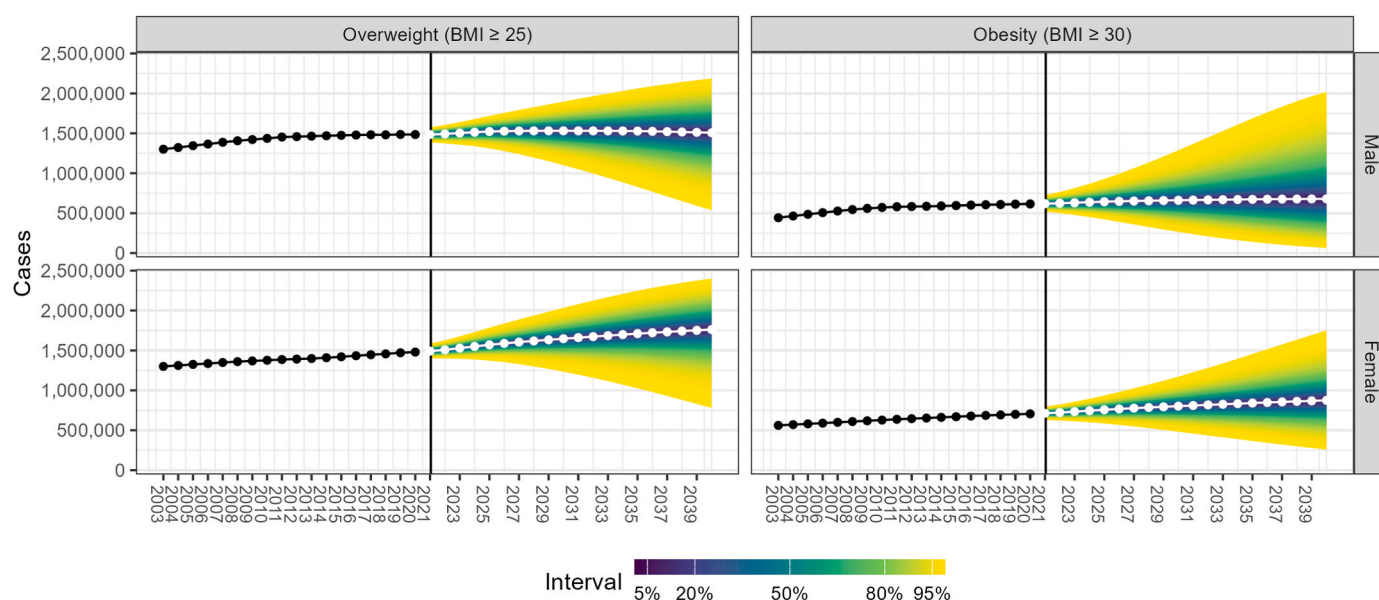


Fig. 1. Historic and projected estimates of the number of prevalent cases of excess weight in Scotland, by sex and study outcome, 2003 to 2040.

though the 2040 estimate remains 1 % higher than in 2019. We estimate a larger increase of 8 % in projected cases of male obesity from 2019 to 2030 with a further increase of 3 % between 2030 and 2040, representing an increase of approximately 66,000 across the full projection period. In 2040, for males we estimate 1,507,405 (95 % CrI: 823,527 to 2,042,332) cases of overweight (64 % of adult males) (appendix p 13) and 678,348 (95 % CrI: 153,768 to 1,661,157) cases of obesity (29 % of adult of males) (appendix p 15).

The projected cases of overweight for females are estimated to increase by 20 % from 2019 to 2040; an increase of approximately 290,000, with over half of this increase (61 %) estimated to occur by 2030. We estimate larger increases of 25 % in the number of female cases of obesity from 2019 to 2040; representing an increase of approximately 175,000, with over half of this increase (58 %) estimated to occur by 2030. In 2040, for females we estimate 1,761,356 (95 % CrI: 1,122,603 to 2,263,378) cases of overweight (68 % of adult females) (appendix p 14) and 874,837 (95 % CrI: 400,863 to 1,470,267) cases of

obesity (34 % of adult females) (appendix p 16).

When we assess historic sex-specific differences in the estimated cases of overweight, there were slightly fewer cases for females than males between 2003 and 2019. However, by 2040 we estimate 17 % more cases of overweight for females compared to males (appendix pp 13–14). For obesity, estimated cases were higher for females than males between 2003 and 2019, although the scale of difference fluctuates across the study period. We estimate 29 % more cases of obesity for females compared with males by 2040 (appendix pp 15–16).

We estimated the probability of scenarios of increased adult excess weight prevalence by 2030 and 2040, separately, for males and females (Fig. 2). For males, there was an estimated probability of 32 % of any increase in the prevalence of overweight by 2030, with a respective probability of 33 % for 2040, compared to the reference. Respective probabilities for obesity were considerably higher (2030: 50 % and 2040: 51 %). The estimated probabilities for females were much higher, compared with males. For females, we estimated there was a 70 %

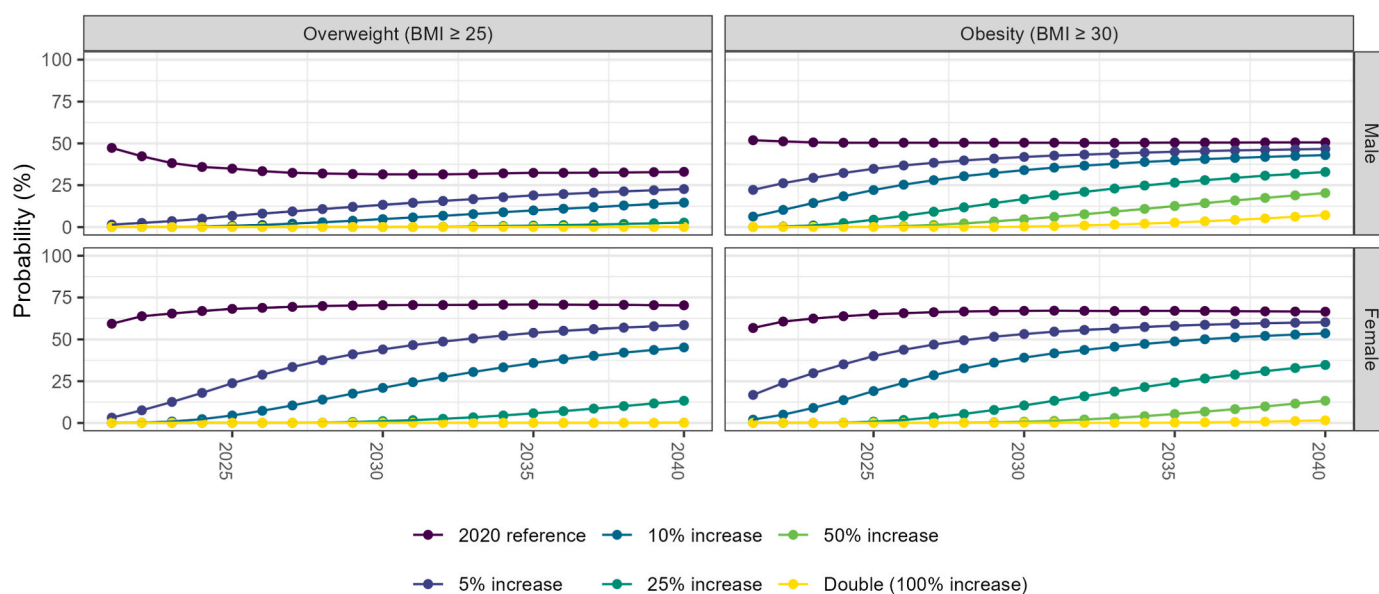


Fig. 2. Probability of an increase in the adult prevalence of excess weight, by sex and study outcome. Based on the marginal posterior distribution of future projections from the Bayesian APC analyses.

probability of an increase in the prevalence of overweight by 2030 and for 2040, with slightly lower respective probabilities for obesity (2030: 67 % and 2040: 67 %).

The historic age structure of cases of excess weight for males and females has changed from 2003 to 2019, a change which is projected to intensify further by 2040 (Fig. 3). The impact of overweight and obesity in males and females aged 65 years and above has become disproportionately larger over time, relative to younger age-groups. Most prevalent cases of excess weight were aged between 35 and 64 years, although there are estimated to be relatively fewer cases in this age-group by 2040 compared to the early study period due to the increasing numbers of prevalent cases aged 65 years and above.

3.4. Model validation and sensitivity analyses

We found that the lowest DIC and WAIC values, and therefore the most appropriate models, were those that included age, period, and cohort effects, as well as the fixed effects of sex and area-level deprivation. This finding was consistent for overweight and obesity models for both males and females (appendix p 5). The explained variability for overweight models was high for both males ($R^2 = 86\%$) and females ($R^2 = 81\%$). For obesity models, the values were slightly lower (Males: $R^2 = 77\%$; Females: $R^2 = 67\%$). When we undertook a sensitivity analysis where we used BMI as the outcome variable, rather than dichotomised outcomes of overweight and obesity, the findings were compatible with that of our main analysis (appendix p 10).

4. Discussion

4.1. Summary of findings

In 2040, we estimate 3.3 million cases of overweight in Scottish adults of which 1.6 million are cases of obesity. Male overweight cases are estimated to continue to increase until 2030 with projections slowly declining after a peak, though the 2040 estimate remains higher than in 2019. For females, over half (61 %) of the increase in overweight cases is estimated to have occurred by 2030, a trend which continues at a reduced rate towards 2040. Estimates were more pronounced for obesity, relative to overweight, with around 66,000 additional male cases and 175,000 female cases projected between 2019 and 2040. With

each consecutive year up 2019 the period effect becomes stronger, adversely influencing an increase in projected cases.

There were slightly fewer pre-pandemic cases of overweight for females than males, however projections in 2040 estimate 17 % more female cases than males. Historically, there have been a higher number of cases of obesity for females, and by 2040 we estimate 29 % more female cases than males. Underpinning these estimates are major shifts in the age distribution of cases. Historic estimates between 2003 and 2019 show that the proportion of male and female excess weight cases aged 65 years and above have increased, a trend projected to further intensify as we move towards 2040.

Our approach allows us to manage expectations for policymakers by communicating the probability of scenario-based increases in cases. We found that by 2040, the probability of an increase in cases of overweight for females was over double that of males (70 % vs 33 %). The probability of an increase in cases of obesity by 2040 was slightly lower than that of overweight but remained higher for females compared to males (67 % vs 51 %).

4.2. Implications for policy and future research

The significant extent of adverse population-level health outcomes attributable to excess weight are clearly evidenced, underscoring the policy-relevance of our findings.^{4,22} Similar patterns whereby female cases were on course to have the largest increases in excess weight were found for Belgium.⁹ Projections of the prevalence of cardiometabolic health conditions in Scotland are also consistent with our results, driven by substantial increases in cases of diabetes which the Global Burden of Disease study highlight as the outcome with the greatest disease burden attributable to excess weight.^{4,23} Furthermore, most studies evidence that increases in obesity are disproportionately higher than overweight which is consistent with our findings.^{24,25} A previous study estimated a slight increase in severe obesity in Scotland by 2035, based on a narrow range of historic data.²⁶

We are clear on the current scale of challenge that excess weight presents.²⁷ Our findings show we are not on the right track; cases of excess weight are projected to increase until 2040, with a greater number of cases of obesity which increases the likelihood of associated health harms. Exacerbating this is the shift towards cases of older ages, and cases developing earlier in the life-course giving rise to an earlier

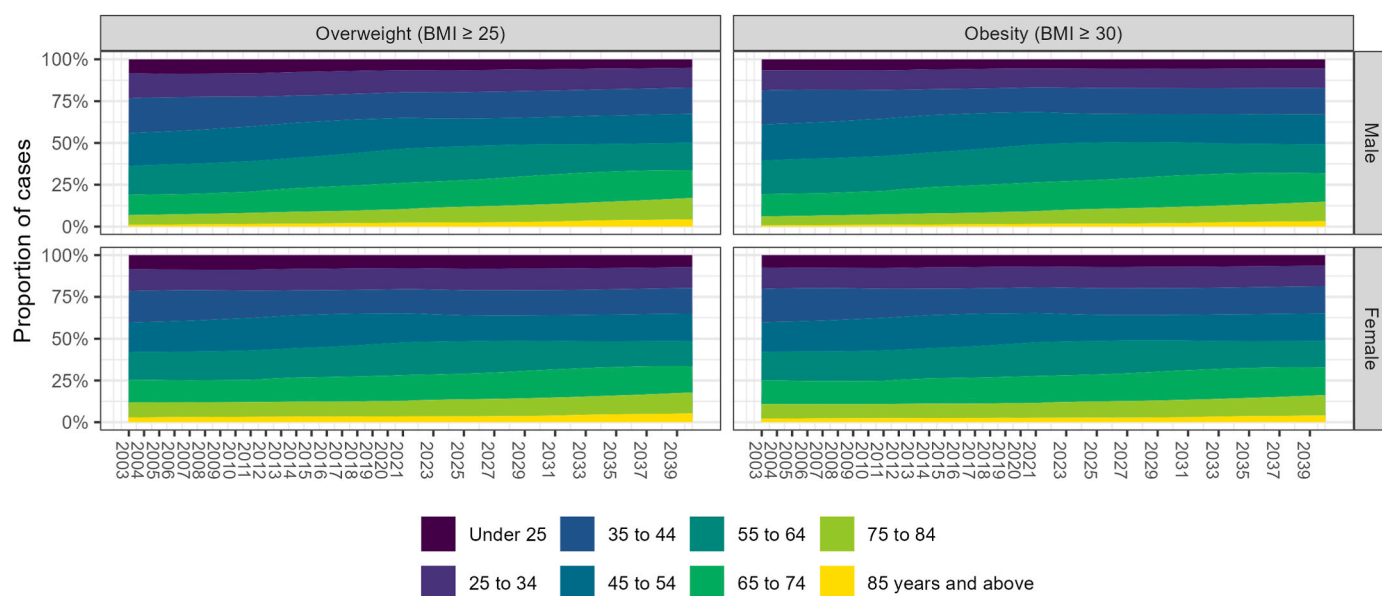


Fig. 3. The age-relative distribution of historic and projected estimates of the number of prevalent cases of excess weight in Scotland, by sex and study outcome, 2003 to 2040.

onset of morbidity. Taken together these present significant challenges because health conditions commonly manifest at the greatest rate at older ages, with multimorbidity becoming increasingly more common, which diminish people's quality of life.^{28,29} There are major implications for the sustainability of publicly funded health and social care services as the projected increase in cases, and shift to an older age-distribution, will likely intensify demand and need for services and treatments to mitigate attributable health conditions manifesting.³⁰ Since excess weight increases the risk of occurrence from a wide range of non-communicable diseases (NCDs), reducing cases would be an effective route to progress towards achieving Sustainable Development Goal (SDG) 3.4 of reducing premature deaths from NCDs by one-third by 2030.³¹ Our findings indicate that any prospect of making progress towards achieving SDG 3.4 by 2030 is likely to be hindered, rather than supported, by trends in excess weight.³²

Recently, the Scottish Public Health System delivered a consensus statement defining key actions to improve diet and weight in Scotland, to generate improvements in population health and wellbeing.³³ Our findings suggest an increasing period effect which shows no evidence of waning, emphasising the substantial role which commercial, economic, and environment factors have in not only influencing current rates, but in intensifying future trends.³⁴ A multi-faceted approach which addresses the food environment through regulation, taxation, product reformulation, and affordability is fundamental to making progress in addressing rates of excess weight and reversing the increasing influence of period effects.³⁵ In addition, availability of weight management and support services, providing evidence-based treatment and support for people living with obesity, will also be crucial.³³ Evidence indicates that around half of the Scottish public would support the taxation, and regulation, of unhealthy food products to tackle excess weight.³⁶

Our study projected the prevalence of excess weight in Scottish adults, so did not consider those aged under 16 years. There are greater uncertainties associated with BMI for children than for adults, as measurements can be influenced by factors such as changes in growth patterns between boys and girls, particularly during puberty when growth is faster, as well as greater contention over which cut-offs to use to define overweight and obesity.³⁷ Estimates for Scotland indicate substantial concerns regarding excess weight in children, with around one-third of those starting primary school being at risk of overweight and those in Scotland's most deprived communities twice as likely than those in the least deprived communities; a widening gap in recent years.³³ We hypothesise the adverse influence of period effects would also be a key consideration for those aged under 16 years. Further research is warranted to establish whether period effects are having an increasing influence in the household and family settings, or whether the chain is being broken and increasingly children and young people are being enabled to get the best start in life, independent of the weight status of other members of their residency.

4.3. Strengths and limitations

Our study used the best available data on BMI in Scotland from 2003 to 2019, based on interviewer-validated height and weight measurements. Measurements were free of biases related to the societal stigma of excess weight, since people tend to self-report lower values.³⁸ There was an absolute difference of 0.97 kg/m² between objectively measured, and self-reported, BMI measurements in Belgium.^{38,39} The repeated cross-sectional nature of our sample allowed us to establish, age, period, and cohort effects related to the prevalence of excess weight, despite having unequal periods.¹⁴

More recent Scottish Health Surveys allow for the sampling of multiple adults per household, removing the need for selection weights, however this design may lead to patterns of weight clustering within households. Adult response rates have decreased over time, which may introduce systematic biases if the prevalence of excess weight is different in non-respondents compared to respondents.^{40,41} To address this

limitation, we utilised survey weights to correct for differences in sampling and response rates. Whilst we do not have evidence directly related to excess weight, evidence suggests alcohol-related health harms in non-respondents are higher than those of respondents.⁴² Since inequalities in excess weight are less pronounced than those which are alcohol-related, we hypothesise that any bias is likely to be lower than have been estimated for alcohol harms. To address the problem of outcome dichotomisation we undertook a sensitivity analysis using BMI as the study outcome and observed trends consistent with our main findings. Our sample was based on data up to 2019 which was the most up-to-date data available in the UK Data Service at time of analyses.

There are inherent limitations to all population projections because they are based on assumptions about future trends rather than on certainties. The impact of these assumptions is apparent for the most recent population projections for Scotland, which projects a higher population by 2047 compared to the previous three sets of projections.¹⁶ The accuracy of our projections is therefore dependent on assumptions made regarding key demographic drivers, such as fertility, mortality, and migration. Unforeseen changes in these components can significantly impact the projected population size and structure. Consequently, projections should be interpreted as plausible scenarios that are subject to a considerable degree of uncertainty. Nevertheless, our projections are based upon the most recently published population estimates and are therefore highly relevant as they reflecting the most up-to-date demographic assumptions.

The main advantage of our study design was that period and cohort effects serve as proxies for events such as risk factors, public health and medical interventions, which are often difficult to measure directly.⁴³ Projections inherently include an increased level of uncertainty as they will be influenced by changes we have not yet experienced, as yet to be experienced events could influence the over- or under-estimation of the actual number of prevalent cases. The key assumption underlying these estimates is that trends in the future will continue based on historical trends, and the validity of this assumption cannot be tested until we are able to collect these data in the future. Nevertheless, public health foresight analysis is an important tool for effective planning of policy and medical interventions development and implementation. Our study utilised HAPC models which allow for a mixture of fixed and random effects, whereas there is a linear dependency between age, period, and cohort effects in classical APC models. The application of Bayesian HAPC models permit the direct interpretation of future estimates in credibility terms, for example estimating the probability of prevalence increases of at least 10 %. This is a strength over applying APC models in a frequentist sense, as frequentist confidence intervals cannot be interpreted in terms of credibility, strengthening the potential of our findings in managing expectations when informing discussions over improving these trends.

We found successive birth cohorts had a reduced risk of excess weight, when isolated from other effects. One explanation supporting these findings could be related to successive birth cohorts being increasingly exposed to the risks posed from excess weight on health, and increased education and awareness over how to reduce the risk of associated harms. This could include the influence of interventions and awareness on more recent cohorts, such as policies in educational institutions. However, we cannot rule out that other factors may be influencing the cohort-specific risk estimates. For example, we have observed that a greater proportion of respondents in more recent surveys are from less deprived areas than previous surveys. This could highlight that respondents are seemingly increasingly benefiting from greater access to factors influencing healthy weight such as healthy food, fitness facilities, and health education, which could reduce the overall cohort-specific risk. Also, given that follow-up for more recent cohorts is limited, compared to those of previous cohorts, we cannot rule out that longer-term analyses may reveal a different pattern later in life, which we are unable to account for, for factors related to delayed onset or changes in the drivers of obesity such as ultra-processed foods. It

remains tricky to interpret the role of birth cohort effects without considering how period, and age, effects are influencing rates of excess weight, as well as the fixed effects. The estimated adversely increasing period effects are an important warning as they indicate that tackling the impact of commercial, economic, and environmental determinants of excess weight are likely to be the most effective means of improving prevalence rates. Since those who have been overweight and living with obesity for long periods of time are at greater risk of premature mortality than those who are not, we hypothesise that this would at least partially explain why we found that age effects reduce for age-groups 65 years and above.

4.4. Conclusion

Left unaddressed, we estimate a substantial increase in adult excess weight in Scotland by 2040, particularly for females, compounded by increases in obesity and in the proportion of older cases. These estimates are not inevitable and underscore the need to accelerate progress towards a multi-faceted prevention-focused approach. Further development of weight management and support services, providing evidence-based treatment and support for people living with obesity, will be important. However, crucially, the key focus must be on improving the food environment through regulation, taxation, product reformulation, and affordability.

Author statements

Author statements

Ethical approval

Ethical approval was not required as this analysis was undertaken on existing, anonymised data. The data that support the findings of this study can be accessed via the UK Data Service –

<https://ukdataservice.ac.uk/>. The UK Data Service provides trusted access and training to use the UK's largest collection of economic, population and social research data for teaching, learning and public benefit.

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Competing interests

RDP and BD are salaried employees of Sciensano. Sciensano were funded by Public Health Scotland to undertake the analyses pertaining to this original research submission. All other authors have no competing interests.

Author contributions

GMAW, RDP, BD, FL and EF generated the idea for the study. GMAW, RDP and BD developed the methodological approach. RDP carried out all analyses and visualisation of results. GMAW coordinated the study and drafted the original manuscript. All other authors provided critical input into the interpretation of the results, revisions to the manuscript, and approved the final draft.

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idea and design of this study, but sadly passed away during its implementation.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.puhe.2025.105981>.

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